# QR SHIELD

Solution Blueprint • CYBER BOYS

## 1. Problem statement

QR codes are convenient, but users cannot always tell where a code leads before scanning or acting on it. A substituted sticker, phishing page, look-alike domain, shortened URL, or unfamiliar payment-related payload can encourage unsafe decisions.

## 2. Proposed solution

QR Shield is a browser-based QR safety companion. It decodes QR content and displays explainable, rule-based indicators so users can pause, inspect the destination, and make a more informed decision. It is a warning aid—not a guarantee, payment-recipient verifier, or live threat-intelligence service.

## 3. Target users

Students and shoppers; UPI and mobile-payment users; small businesses, campuses, cafés, and event organizers. These are proposed user groups and should be validated through interviews and pilot testing.

## 4. Key prototype features

| Feature | Current presentation-described behavior |
|---|---|
| QR input | Camera scanner, QR-image upload, and pasted content; availability depends on browser/device support. |
| Rule-based URL review | SAFE / CAUTION / RISK heuristic score with findings for selected suspicious patterns. |
| Payload inspection | Detects URL, Wi-Fi, contact, UPI, phone, email, and other payload types. |
| Alerts and history | Sound alert for high-risk results and browser-based scan history. |
| Reports and exports | QR generation, metadata inspection, SHA-256 fingerprint, CSV/JSON export. |
| MR.Shield guide | Built-in rule-based guidance; not connected to a live AI model. |

## 5. User workflow

Scan / upload / paste QR content → decode payload → classify payload type → apply local rules → show score, findings, and guidance → user decides whether to continue → optionally review or export history.

## 6. Business model concept

The presentation proposes a free tier at ■0, Pro at ■99/month, and Pro+ at ■199/month. These are concept prices, not validated revenue. Subscription controls in the prototype are demo-only and do not process payments. Conduct willingness-to-pay testing before launch.

## 7. Expected impact

Encourage users to pause before opening unfamiliar destinations; make selected warning signals easier to understand; support QR-safety awareness sessions for campuses and small merchants; and provide a simple record/export workflow for review.

## 8. Safety, privacy, and limitations

Analysis is described as browser-side. History uses browser storage and can be cleared/exported. The app does not provide a definitive safety verdict, live reputation lookup, or final redirect inspection. Local heuristics can miss new scams or flag legitimate URLs. Users should never enter an OTP, password, or payment PIN based solely on a QR Shield result.

## 9. Validation plan

First 30 days: interview users, run usability tests, create a curated test set, and fix accessibility/camera issues. By 60 days: add tests, assess redirect-analysis options, and measure classification errors. By 90 days: pilot with a campus or merchant group and evaluate comprehension, privacy, repeat use, and pricing.

## 10. Success measures to collect

User understanding of warnings; completion rate for scan workflow; false-positive and false-negative rates on a documented test set; time to understand a result; repeat usage; and pilot feedback. No benchmark results were supplied with the presentation.

Source boundary: this blueprint is based on the supplied presentation. Claims about implementation should be checked against the actual source code and tested prototype before submission.